



Voice Synthesis: Replicating the Human Voice with Machine Learning

A technical exploration of vocal model training. We'll compare personalized approaches and online tools.



Introduction to Speech Synthesis

Text-to-speech (TTS) technology converts written text into speech. It's crucial for AI, accessibility, and voice assistants.

TTS Definition

Convert written text into realistic speech.

Importance in AI

Fundamental for natural human-machine interactions.

Common Applications

Voice assistants, audiobooks, digital accessibility.

Project Objectives

The goal is to train a personalized voice model from audio samples.



Sound Quality

Evaluate the clarity and naturalness of the generated voice.



Vocal Fidelity

Replicate the original voice with high timbral accuracy.



Applicability

Test the effectiveness in various practical use scenarios.



Technical Implementation

Language & Libraries

- Python was the main language.
- Libraries: TTS (Coqui), PyTorch, NumPy.



Datasets & Models

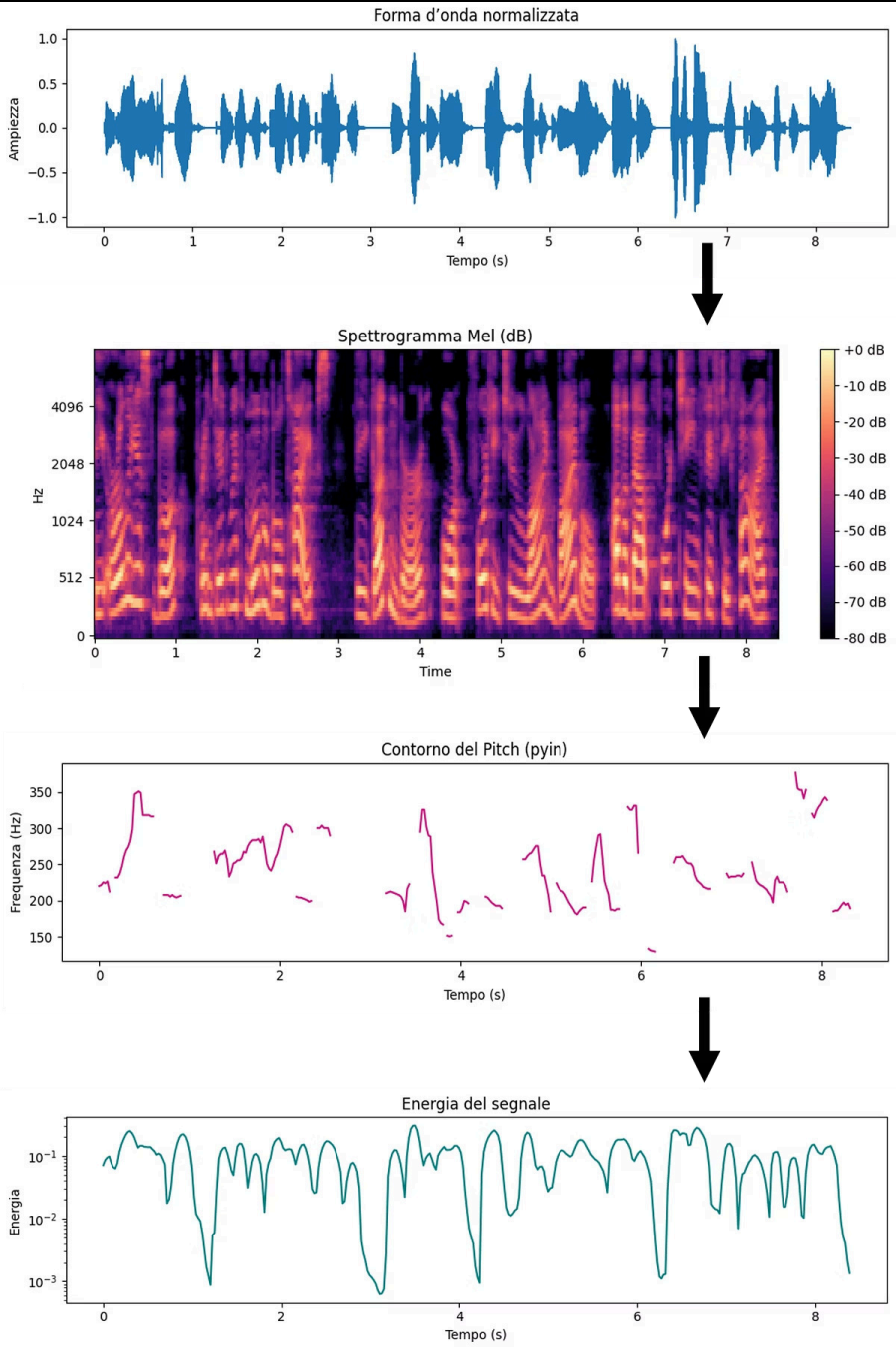
- Fine-tuning on VCTK with LJSpeech.
- Zero-shot TTS: Using Tortoise and XTTSv2.



Environment

- Google Colab with Tesla T4 GPU.
- Suitable environment for training and inference.





Preparation and Pre-data processing

Dataset Collection

Clean voices, high quality audio for training.

Audio Normalization

Sample rate and volume standardization.

Text Transcription

Accurately align text with audio samples.

Feature Extraction

Generation of mel, pitch, and energy spectrograms.

Model Training Process

1

Main Challenges

Limited memory on Colab and long training times.

2

Time & Hardware

Google Colab training with Tesla T4 GPUs. Average time: ~1-2 hours (partial fine-tuning).

3

Optimization

Pre-trained VCTK model, careful choice of parameters (learning rate, scheduler).





Audio Demo and Quality Comparison

Listen to the differences and quality of the text-to-speech.



Trained Model

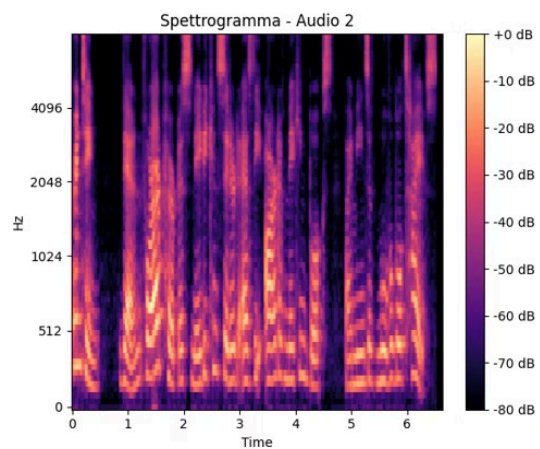
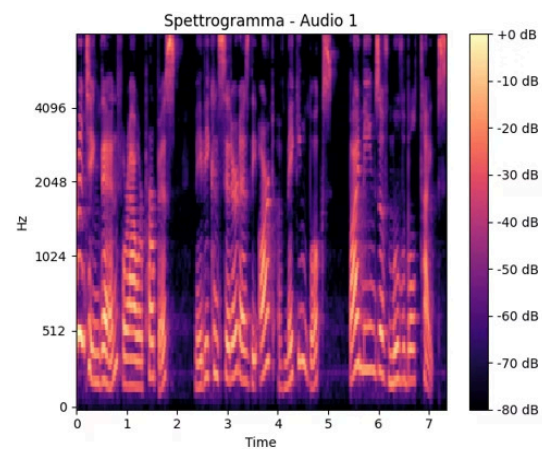
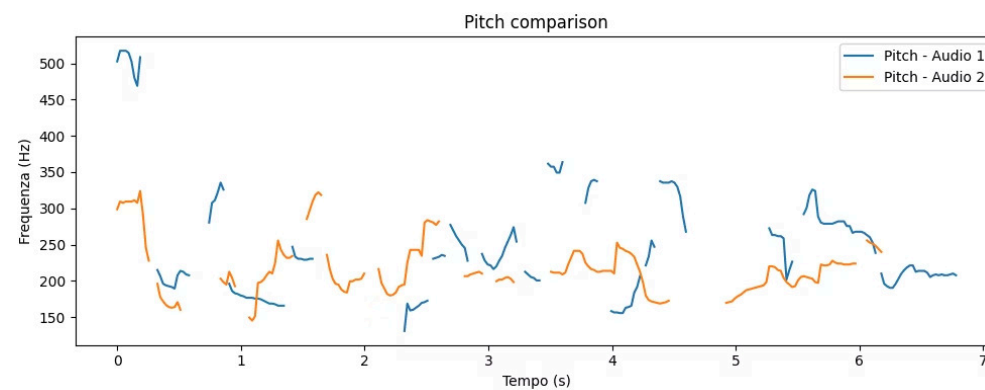
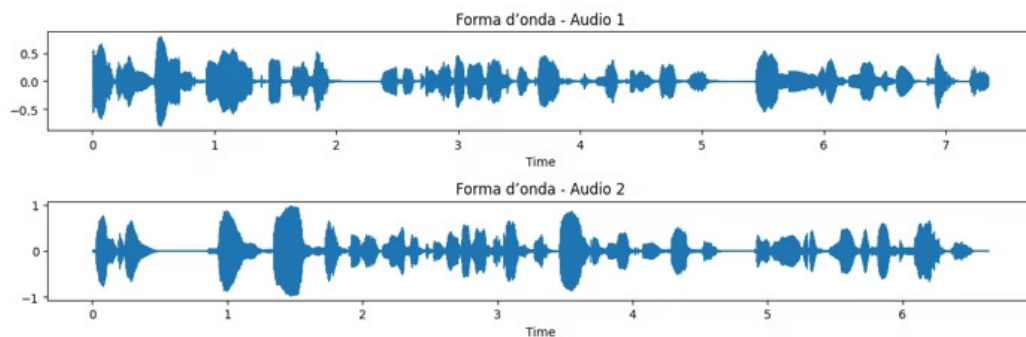
Voice replicated with superior fluidity and intonation.



Online Tool

Acceptable quality, but with less customization.

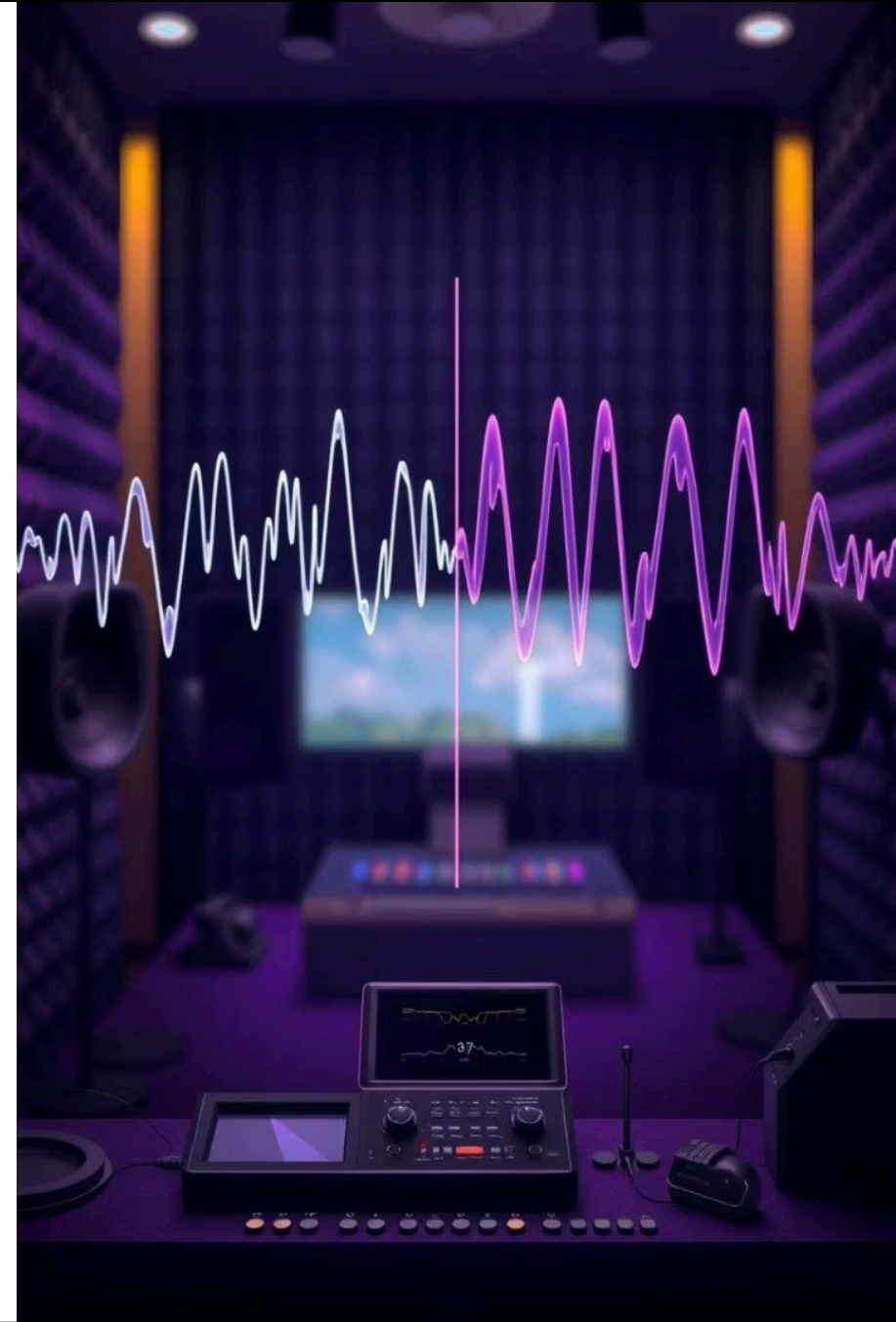
Evaluation of Metrics

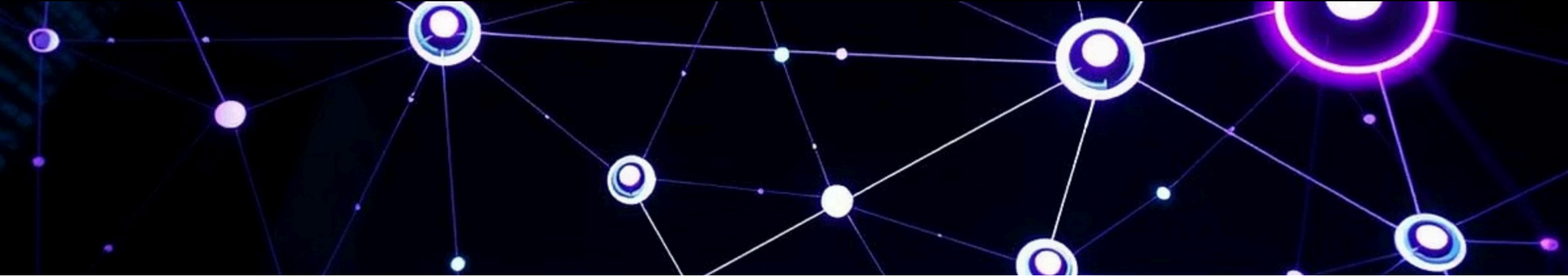


Evaluation of Results

Direct comparison between the trained model and online tools. Online tools offer greater naturalness.

I wait	Python template	Online Model
Comprehensibility	 &	 &
Vocal Fidelity	 &&	 &&&
Naturalness	 &&	
Spectrogram (visual)	[7]	[7]
Realistic pitch	Medium	High





Conclusions and Future Work

It has been possible to synthesize human speech with good quality using pre-trained models.

There is a trade-off between sound quality and voice control.

Proven Feasibility

Custom text-to-speech with Python.

Future Improvements

Longer training, more complex networks, app integration.

Link to Notebooks

[https://colab.research.google.com /
drive/1eP5-
XFkiXycHmkmLT3ZovorK6_bakeIk ?
usp=sharing](https://colab.research.google.com/drive/1eP5-XFkiXycHmkmLT3ZovorK6_bakeIk?usp=sharing)



Andrea Lo Giudice
University of Catania

